

Client spec cards (one per team)

Hand each team one client card. Every design must meet ALL four of its client's numbers. Slope is written as rise:run, so 1:12 means 12 cm of ramp length for every 1 cm of height. Ramp length = rise × the run number.

Client A Community school

Rise to cover	10 cm
Max slope	1:12
Load to carry	a 200 g cart
Portability	folds to fit inside a backpack

Client B Public library

Rise to cover	8 cm
Max slope	1:14
Load to carry	a 300 g cart
Portability	one student can carry it alone

Client C Health clinic

Rise to cover	12 cm
Max slope	1:12
Load to carry	a 250 g cart
Portability	sets up in under one minute

Client D Science museum

Rise to cover	15 cm
Max slope	1:16
Load to carry	a 200 g cart
Portability	the ramp itself weighs under 200 g

Length check (for the instructor): Client A: 10 cm at 1:12 needs about 120 cm of ramp; Client B: 8 cm at 1:14 needs about 112 cm of ramp; Client C: 12 cm at 1:12 needs about 144 cm of ramp; Client D: 15 cm at 1:16 needs about 240 cm of ramp.